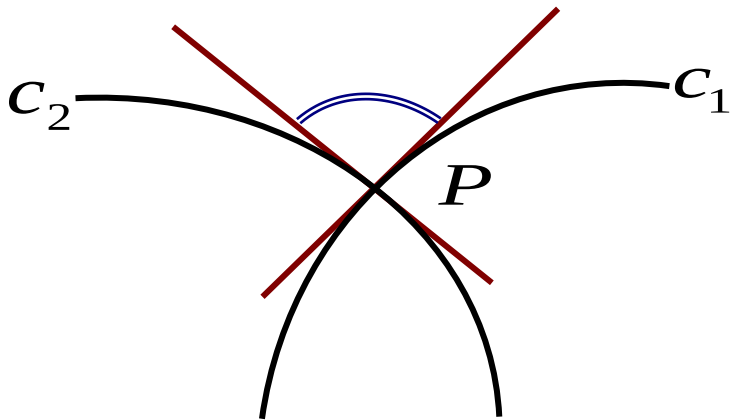


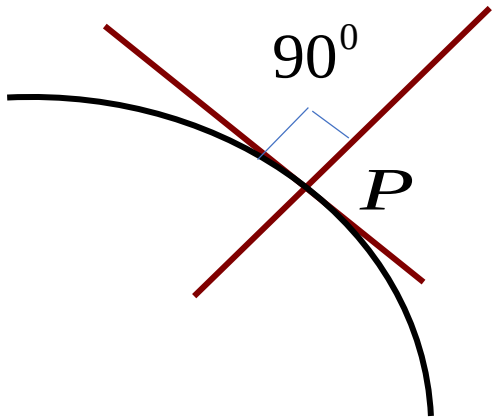
Orthogonal Trajectories

Angle between two curves

If two curves intersect at the point P then the acute angle between the tangents drawn to the curves, is called the **angle of intersection of the curves** at the point P .



For two curves intersecting at the point P , the tangents drawn to the curves at the point P , is perpendicular then it is called that the curves intersect **orthogonally** at the point P .

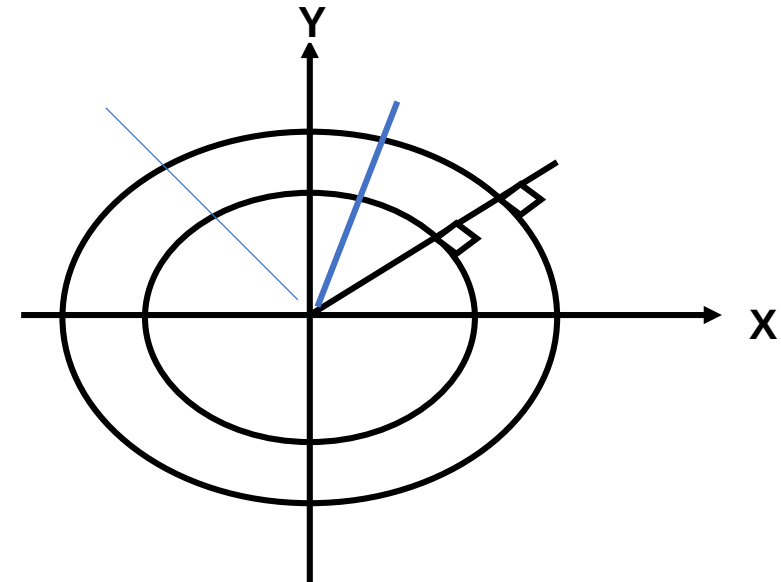


Example

$$x^2 + y^2 = C \rightarrow (1)$$

$$y = Ax \rightarrow (2)$$

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Procedure to find orthogonal trajectory:

Let $F(x, y, c) = 0$ be the equation of the given family of curves,
 c -arbitrary parameter.

- Differentiate $F(x, y, c) = 0$ w.r.t. x and eliminate c .
- Replace $\frac{dy}{dx}$ by $-\frac{1}{dy/dx}$ in the above differential equation.

This will give the differential equation of the orthogonal trajectories.

- By solving this differential equation, we get the required orthogonal trajectories.



Find the orthogonal trajectories of the family of parabolas $y = c x^2$

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Differentiate w.r.t. x

$$\frac{dy}{dx} = 2cx$$

Eliminate C .

$$c = \frac{y}{x^2}$$

$$\frac{dy}{dx} = \frac{2y}{x}$$

Replace $\frac{dy}{dx}$ by $-\frac{1}{dy/dx}$

$$-\frac{1}{dy/dx} = \frac{2y}{x}$$

$$-\frac{dx}{dy} = \frac{2y}{x}$$

Equation of orthogonal trajectories

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$$-\frac{dx}{dy} = \frac{2y}{x}$$

$$x \, dx + 2y \, dy = 0$$

$$\frac{x^2}{2} + y^2 = a$$

$$\frac{x^2}{(\sqrt{2a})^2} + \frac{y^2}{(\sqrt{a})^2} = 1 \rightarrow (1)$$

major axes $\sqrt{2a}$

semi-minor axes $\sqrt{a}$

here $a > 0$ is an arbitrary constant. (1) gives a family of ellipses.

